

Date	Team ID	Project Name
04 JULY 2026	SWTID-2026-2662	AI Powered Debt Relief & Financial Recovery Platform

AI Powered Debt Relief & Financial Recovery Platform

An AI-powered web application that helps borrowers manage debt, analyze financial health, and generate professional settlement/negotiation strategies — built with **React.js**, **FastAPI**, **SQLite**, **SQLAlchemy ORM**, and **Google Gemini API**.

📌 Overview

The AI Powered Debt Relief & Financial Recovery Platform simplifies and automates debt management for borrowers. Users can record loan details, analyze their financial health, generate AI-powered negotiation strategies, and receive personalized settlement recommendations — all within a secure, modern digital environment.

✨ Core Features

- **AI-Powered Settlement Recommendation** - Analyzes loan and income details to generate personalized settlement suggestions and debt stress analysis.
- **Intelligent Negotiation Letter Generation** - Uses Google Gemini AI to draft professional, lender-specific negotiation emails and settlement letters.
- **Financial Health Tracking & Loan Analysis** - Dashboard for monthly surplus, EMI ratio, debt stress level, and settlement percentages, with a history of AI-generated negotiations.

🔧 Tech Stack

Layer	Technology
Frontend	Streamlit / React.js (Vite)
Backend	FastAPI (Uvicorn)
Database	SQLite with SQLAlchemy ORM
AI Engine	Google Gemini API (<code>gemini-1.5-flash</code>)
Auth	JWT (python-jose) + Werkzeug password hashing

Layer	Technology
Deployment	Docker, Render / AWS
Version Control	Git & GitHub, GitHub Actions (CI/CD)

 Project Structure

```
FinRelief-AI/
├── app/
│   ├── main.py
│   ├── database.py
│   ├── auth.py
│   ├── models/
│   ├── schemas/
│   ├── routers/
│   └── services/
├── streamlit_app/
│   ├── app.py
│   ├── api_client.py
│   ├── components.py
│   ├── styles.css
│   └── pages/
├── requirements.txt
├── runtime.txt
└── .env
```

 Getting Started

Prerequisites

- Python 3.11+
- Node.js (for React/Vite frontend, if used)
- Google Gemini API key

Installation

```
bash
```

```
# Clone the repository
git clone https://github.com/nakkahemasaisree-cmd/AI-powered-debt-relief-and-fi
cd AI-powered-debt-relief-and-financial-recovery-platform

# Install backend dependencies
pip install -r requirements.txt

# Configure environment variables
cp .env.example .env
# Add your GOOGLE_API_KEY, SECRET_KEY, DATABASE_URL

# Run the FastAPI backend
uvicorn app.main:app --reload

# Run the Streamlit frontend (in a separate terminal)
streamlit run streamlit_app/app.py
```

 Modules

Module	Description
Authentication	Secure registration, login, JWT-based session management
Loan Management	Add, view, and delete loan records
Financial Engine	Computes EMI ratio, debt-to-income ratio, surplus, stress level
Settlement Engine	Rule-based settlement percentage & risk scoring
AI Engine	Gemini-powered negotiation strategy & letter generation (with rule-based fallback)
Dashboard	Financial health visualizations and loan prioritization

 Testing

- **Functional Testing:** Registration/login, loan input validation, AI generation, API connectivity — all passed.

- **Performance Testing:** Conducted using Apache JMeter; average response time 1.42s, throughput 55 req/sec, error rate 0.30%.
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Skills & Domain

Python FastAPI SQLite Generative AI Responsible AI LLMs Prompt Engineering
Gemini AI API Integration Cloud Computing

Domain: Google Cloud Generative AI

Team

Name	Role
Hema Sai Sree	Team Lead & Backend Developer
Nithya Sree	Project Coordinator & Frontend/UI Developer
Dharani	AI & Financial Processing Developer
Shalini	Backend Developer & Security

Future Scope

- Real-time banking & credit bureau API integration
 - Migration to PostgreSQL/MySQL with cloud deployment
 - Voice-enabled chatbot & multilingual negotiation support
 - Multi-Factor Authentication (MFA) and role-based access control
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Disclaimer

This project is an academic/internship proof-of-concept. It does not connect to real banking systems and should not be used as a substitute for professional financial or legal advice.

Links

- **GitHub Repository:** <https://github.com/nakkahemasaisree-cmd/AI-powered-debt-relief-and-financial-recovery-platform>

- **Demo Video:** <https://drive.google.com/file/d/1AoQy2YSE4mAHeExVLR-KgtKG2Yc36TBm/view?usp=sharing>